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Posttraumatic epilepsy: Integrating clinical, inflammatory, and genetic profiles in traumatic brain injury patients☆

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ABSTRACT

Objective: This study aims to assess the clinical, inflammatory, and genetic profiles of traumatic brain injury (TBI) patients over a 2-year follow-up period, focusing on the development of posttraumatic epilepsy (PTE).
Methods: Fifty-nine patients with acute TBI were recruited in the emergency unit of a hospital in Brazil. Clinical data and blood samples were collected after 10 days of hospitalization for posterior genetic profile (Apolipoprotein E- ApoE and Glutamic Acid Descarboxylase-GAD sequencing) analyses. A subset of 19 patients were assessed for cytokine markers (mRNA expression). The development of PTE was investigated for two years following TBI. Statistical analyses including univariate analysis, multiple correspondence analysis, and Mann-Whitney test were performed.
Results: Analysis revealed an association between severe TBI and requirement for neurosurgery and polytrauma ($p<0.05$), as well as the development of PTE over a two-year follow-up period ($p<0.05$). Multiple correspondence analysis identified two distinct profiles associated with PTE and Non-PTE outcomes. The PTE profile showed a higher prevalence of the ApoE genotype E3/E3 and GAD1 SNP (rs769391) genotype AA in our study, while the Non-PTE profile showed a higher presence of E3/E4. mRNA expression analysis demonstrated acute elevated levels of TNF-α in the PTE group as compared to Non-PTE patients (6.70 ± 1.53 vs $5.31\ \pm0.33$, $p<0.01$).
Significance: Our findings underscore the multifactorial nature of aspects potentially contributing to PTE. It is unlikely that any single factor might in isolation have a strong causative influence over the development of epilepsy after TBI. Our results provide a suggestion of potential clustering that might be relevant as prognostic factors for PTE.

1. Introduction

Traumatic brain injury (TBI) is a public health problem that affects nearly 50 million people every year and stands as a leading cause of death and enduring disability (Maas et al., 2017). The impact of external forces on the head results in anatomical and functional brain damage that can lead to posttraumatic epilepsy (PTE), which affects approximately 5 % of all focal epilepsy cases (Agrawal et al., 2006). The risk of developing epilepsy is proportional to the severity of the trauma, ranging from 3 % for mild injury to 16 % for focal cerebral injury (Karlander et al., 2021). Despite years of research, the relationship between TBI and epilepsy is still not fully understood.

Nowadays, there is no cure or treatment to effectively prevent the development of PTE after a brain injury (French et al., 2021). The mechanisms underlying the emergence of epilepsy in these patients are still unknown. Despite extensive research on the imbalance between

☆ We confirm that we have read the Journal's position on issues involved in ethical publication and affirm that this report is consistent with those guidelines.
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glutamatergic and GABAergic systems implicated in seizures and epilepsy development, conclusions are limited by isolated findings and the lack of understanding due to intricate interactions (Sarlo and Holton, 2021).

The inflammatory response in the brain tissue is one of the principal consequences triggered by the TBI that is associated with the development of PTE. This response is initiated and regulated by various cellular molecular mechanisms, such as cytokines expression that are rapidly released in response to TBI (Diamond et al., 2015; Sharma et al., 2019; Mukherjee et al., 2020). Some cytokines can be pro-inflammatory, propagating the inflammatory response, while others can be anti-inflammatory, inhibiting the inflammatory response (Pitkänen and Lukasiuk, 2011; Vezzani et al., 2019; Mukherjee et al., 2021; Woodcock and Morganti-Kossmann, 2013; Webster et al., 2017).

In addition, recent molecular studies, have provided insight into genetic profiles that may contribute to the formation of ictogenic neuronal networks. These include biomarkers consisting of genes regulating synaptic or neuronal plasticity —including specific polymorphisms of apolipoprotein E (ApoE) and glutamic acid decarboxylase (GAD). ApoE gene is involved in making a protein that helps carrying in brain lipid regulation, synaptic plasticity, and neuronal membrane maintenance, plays a crucial role in neuronal tissue repair and regrowth following neurotrauma. With three different alleles (ε2, ε3, ε4), the phenotype–genotype relationship of the ApoE gene has been associated with diverse pathological conditions and cognitive traits (Singh et al., 2006; Merritt et al., 2021).

Another gene that seems to contribute to PTE is GAD1. This gene coded GAD67, is one of the two forms of the enzyme GAD in charge of synthesizing GABA from glutamate in the brain (Cotter et al., 2017). GAD1 has three single nucleotide polymorphisms (SNPs), rs3828275, rs3791878, and rs769391, associated with the occurrence of post-traumatic seizures (Darrah et al., 2013). The rs3828275 is correlated with a predisposition to present early seizures, while rs3791878 and rs769391 are associated with posttraumatic seizures in a time interval of 1–6 weeks after trauma (Cotter et al., 2017).

Therefore, our study aims to integratively evaluate the clinical, inflammatory, and genetic profiles of TBI patients over a 2-year period concerning the development of PTE.

2. Materials and methods

2.1. Participants and ethical considerations

The comprehensive analysis of genetic, clinical, and inflammatory profiles related to PTE was conducted in light of an ongoing phase III clinical trial assessing the efficacy and safety of biperiden, a muscarinic acetylcholine receptor antagonist, to mitigate the risk of PTE development in adults (clinicaltrial.gov: NCT01048138; (Foresti et al., 2023)).

The study received approval from the Research Ethics Committee of the Universidade Federal de São Paulo (CAAE: 08533513.6.1001.5505) and from HCFMUSP (CAAE: 08533513.6.2002.0068). According to national regulations and ethics committee requirements, written informed consent was obtained from the patient or a legally acceptable representative.

Following the clinical-trial protocol, patients with acute TBI were consecutively enrolled in the emergency unit of the Hospital das Clínicas, Faculdade de Medicina at Universidade de São Paulo (HCFMUSP) from 2018 to 2020. After inclusion, the participants were randomly assigned to receive the protocol intervention (see Foresti et al., 2023 for additional details). Inclusion criteria comprised the following: 1. Both sexes; 2. Age between 18 and 75 years; 3. Acute TBI; 4. The presence of contusion and/or acute intraparenchymal hemorrhage indicated in computed tomography (CT); 5. signed informed consent (possibly by a relative). Exclusion criteria included: 1. History of seizures or previous epilepsy; 2. Presence of risk factors to develop epilepsy (e.g., perinatal lesions, febrile seizures, meningitis, encephalitis, use of anti-seizure drugs, medical history of neoplasia, severe comorbidities, neurodegenerative disease, stroke, and cognitive dysfunction); 3. Pregnancy; 4. Patients at increased risk of developing adverse reactions to biperiden such as cardiac arrhythmias and atrioventricular block, glaucoma, and benign prostatic hyperplasia (as for clinical trial); 5. Patients currently involved in other clinical trials.

2.2. Clinical assessment and follow-up data

Clinical data, including age, sex, cause of TBI, presence of polytrauma, the necessity of neurosurgery, and occurrence of acute symptomatic seizures (ASS) were collected during the hospitalization period. The ASS refers to immediate seizures occurring within 24 hours of brain insult or early seizures occurring 1–7 days after the precipitating injury. These seizures are not considered indicative of epilepsy. It is imperative to clarify that our emergency unit lacks the capability for continuous EEG and video monitoring, potentially resulting in subtle acute seizures that might have not been reported. Blood samples were collected within 9–11 days after TBI for inflammatory and genetic profiles of TBI patients.

After hospital discharge, patients underwent a 2-year follow-up by direct attendance by the patients to the hospital clinic whenever possible and by telephone interview performed by the staff physicians in the instances where COVID restrictions were under effect to investigate the development of PTE, characterized by spontaneous recurrent seizures occurring 7 days or more after the initial brain injury. Clinical evaluation was performed by blinded experienced epileptologists. Therefore, the PTE diagnosis was defined based on detailed clinical history that was obtained from the information of the patient and family members (Fisher et al., 2017; Chen & Koubeissi, 2019).

Other outcome measures of the clinical trial, such as electrographic recordings, cognitive deficits, and functional assessments, undergo additional evaluation beyond the scope of this report.

2.3. ApoE and GAD sequencing

2.3.1. DNA extraction and Polymerase Chain Reaction (PCR)

Peripheral blood samples were collected into EDTA tubes. Genomic DNA (gDNA) was extracted using Chellex100 resin (BioRad,ref1421253, Hercules, USA), following the Walsh and colleagues (2013) protocol (Walsh, Metzger and Higuchi, 2013). Post-extraction, gDNA concentration, and purity were assessed using a NanoDrop® spectrophotometer (ND-1000, UniscienceTM).

ApoE genotyping involving specific alleles on two SNPs (rs429358 and rs7412), and GAD sequencing utilizing three SNPs (rs3791878, rs3828275, and rs769391) were performed. PCR was carried out with specific forward and reverse primers (Exxtend Biotechnology, São Paulo, Brazil). Detailed information in Supplement Material and Table S1.

2.3.2. Sanger sequencing

Each SNP was sequenced using BigDye Terminator v3.1 cycle sequencing kit at the ABI Prism 3500xl Genetic Analyzer sequencer. Data analysis was performed with Geneious 10.2.3 software (https://www.geneious.com), and sequences were compared with the DNA reference sequence (ApoE: NG_007084) (GAD: NG_021477).

2.4. mRNA expression of inflammatory cytokines

2.4.1. RNA isolation and cDNA synthesis

Total RNA was extracted from peripheral blood samples using Tempus™ blood RNA tubes (Applied Biosystems) and isolated with the Tempus™ Spin RNA Isolation kit (Applied Biosystems), following the manufacturer's protocol. Reverse transcription of RNA samples was carried out using the High-Capacity cDNA Reverse Transcription Kit (Applied Biosystems, California, USA) with amplification conditions set

at 25 °C for 10 min, 37° for 120 min, and 85°C for 5 min.

2.4.2. mRNA expression - Real-Time PCR

Quantitative PCR, utilizing TaqMan Universal PCR Master Mix (Thermo Fisher Scientific), was performed for five genes of interest: IL1-β, IL-6, IL-10, TNF-α, and CCL2 (see Table S2) and GAPDH as the reference gene. See detailed information in Supplemental Material.

2.5. Statistic analysis

Data was tested for normality and homogeneity. Continuous variables were presented as mean ± standard deviation, while categorical variables were described as distribution and frequency. To examine the associations between individual variables and PTE and TBI severity, Chi-square and Fisher's exact test were used, risk ratios (RR) or odds ratio (OR) with corresponding 95 % confidence interval (CI) were reported.

For a comprehensive understanding of relationships among categorical variables, we conducted a multiple correspondence analysis (MCA). MCA is used to analyze a set of observations described by a set of nominal variables distributed in an Euclidean space and helps to describe patterns of relationships using geometrical methods (Ayele, Zewotir and Mwambi, 2014).Data points are represented as coordinates in this space, and its interpretation is based upon proximities between points in a simplified map, usually two or three dimensions. The graphical representation facilitates the comprehension of associations between variables. While primarily an exploratory technique, MCA can reveal meaningful groupings of variable categories in dimensional spaces, offering crucial insights into category relationships (Costa et al., 2013).

In our study, we used MCA to speculate whether PTE would be linked with more severe outcomes.

This includes various factors like: PTE, Glasgow Coma Scale (GSC) at the hospital admission, localization of the lesions (Frontal, Temporal, Parietal, Occipital), and others (refer to Tables 1 and 2).

For mRNA expression analyses, outliers were identified with the ROUT method implemented in GraphPad Prism 8 and excluded from the analysis. Group differences were assessed using the non-parametric Mann-Whitney test.

All statistical analyses were performed using Prism software (version 8.0.2.; GraphPad, USA) and the SPSS statistical package (version 29, IBM SPSS Statistics, Armonk, NY, INC). Statistical significance was set at $p<0.05$. Power calculations were performed when significant differences were detected for both quantitative and qualitative data (alpha = 0.05).

3. Results

3.1. Demographic and clinical characteristics of patients with TBI

From seventy-five (n=75) adult patients with acute TBI assessed for eligibility, fifty-nine (n=59) patients were recruited and had blood samples collected and available for genetic analysis. Due to resource and logistic constraints, a smaller number of these patients were assessed for cytokine analyses through mRNA (n=19). A breakdown of the population based on PTE status is provided in Table 1, which contains baseline characteristics such as demographic, clinical, and genetic features for each group.

Most patients were male (79.7 %) and under 45 years (61 %) at the time of TBI. Based on GSC analysis during admission, most of the patients had moderate or severe TBI (66.1 %) (Table 1). Most patients had no family history of epilepsy (59.3 %), neither presented ASS (78 %) during the hospitalization period. Related to brain injury, the majority had just one lesion (46 %) with no midline shift (76 %). Over the 2-year follow-up, 7 (12 %) patients died (see Table 1).

The distribution of ApoE alleles and GAD 1 SNP also is presented in Table 1. In most patients, 61 % had the allele ε3 in homozygosis, while only 5 % presented the ε4 allele in homozygosis. Of the 59 patients, n=20 (33.9 %) presented at least one ε4 allele for ApoE (Table 1). Regarding GAD 1 SNP analysis, 63 % of patients had the rs3791878 wild-type (GG), 51 % had CT in rs3828275, and 53 % had rs769391 wild-type (AA) (Table 1).

As expected, moderate or severe TBI was associated with the presence of polytrauma ($p<0.05$) and the requirement for neurosurgery ($p<0.05$) (Table 2). A total of eight (n=8; 15.4 %) patients developed PTE (Table 2) over a two-year follow-up. The development of PTE was influenced by trauma severity ($p<0.05$) (Table 2).

Additional baseline characteristics, categorized by placebo or biperiden groups, are presented in the Supplementary Material (Table S3). The corresponding results are depicted in Figures S1A-C and S2. Overall, these supplementary analyses indicate that there were no significant differences among the treatment groups based on the original data from the clinical trial.

3.2. Evaluation of SNPs – multiple correspondence analysis

Data of baseline characteristics of the total population (n=59, Table 1) were used for MCA analysis. The results of the MCA analysis suggested a two-dimension MCA solution as the most appropriate for the present study (Fig. 1) in which the first dimension accounted for 13.71 % of the variance, while the second explained 11.12 %, resulting in a total variance of 24.8 %. The criteria for choosing a two-dimension MCA was based on: 1-eigenvalue, values above one were prioritized for greater variability explained. 2- Cronbach's alpha score, higher values show greater reliability of the analysis, and 3- easy interpretation (Cronbach, 1951; Khangar and Kamalja, 2017; Baki et al., 2022). In our analysis, the first and second dimensions showed eigenvalues of 3.29 and 2.68, inertias of 0.137 and 0.112, and Cronbach's alpha values of 0.726 and 0.655, respectively. While the generally accepted lower limit for Cronbach's alpha is 0.70, a slightly lower value is considered acceptable in exploratory research (Johnson and Wichern, 2007). A reduced alpha score in exploratory studies may be attributed to heterogeneous constructs.

Discrimination measures (Table S4) and a joint plot of category points were obtained (Fig. 1). All discrimination measures were below 0.5, with a maximum value of 0.370 (Lesions) for the first dimension and 0.409 (GAD SNP (rs3791878)) for the second dimension.

Based on MCA analysis, we delineated two profiles – one in orange, which we named PTE, and the other in blue, which we named Non-PTE, we assigned these names as we observed that the PTE variable clustered with certain variables, while the Non-PTE variable clustered with others. In the PTE profile, we observed associations with more severe TBI (as per GCS), a higher incidence of polytrauma, an increased necessity of neurosurgery, and elevated mortality. Concerning brain trauma, patients with this profile presented subdural and epidural hematomas and subarachnoid hemorrhage. Brain lesions were characterized by bilateral lesions and lesions mostly located in the temporal and parietal regions. This profile showed a higher prevalence of the ApoE genotype E3/E3 and GAD1 SNP (rs769391) genotype AA in our study.

In contrast, in the Non-PTE profile, we observed moderate and mild TBI, the absence of subdural and epidural hematomas, no fractures, and no subarachnoid hemorrhage. This profile involved one or two lesions, mostly located in the frontal region, with a higher presence of the ApoE genotype E3/E4.

3.3. mRNA expression levels analysis

The mRNA expression levels of IL1-β, IL-6, IL-10, TNF-α, and CCL2 were assessed in the blood samples of 19 patients (Fig. 2). When comparing mRNA expression between the PTE (n=4) and the Non-PTE (n=15) groups, a significant difference was observed only for TNF-α ($p<0.01$). Specifically, patients who developed PTE displayed higher TNF-α mRNA expression (6.70± 1.53) compared to the Non-PTE

Table 1
Demographic and clinical characteristics of patients with traumatic brain injury (TBI).

Variables	Total population (n= 59)	Genetic Population (n=59)			Cytokine Population mRNA (n=19)		
		No-PTE (n=51)	PTE (n=8)	p-value	No-PTE (n=15)	PTE (n=4)	p-value
Sex							
Male n (%)	47 (79.7)	41 (80.39)	6 (75)	0.66	14 (87.5)	2 (66.7)	0.42
Female n (%)	12 (20.3)	10(19.61)	2 (25)		2 (12.5)	1 (33.3)	
Age							
(population age mean ± s.d)	42.5 ± 16.38	43.47±16.58	37±14.76		35.25 ± 11.13	44.33 ± 23.46	
18–30 n (%) (Reference)	16 (27)	13 (25)	3 (38)		6 (37.5)	1(33.3)	
31–45 n (%)	20 (34)	16 (31)	4 (50)	>0.99	8 (50)	1 (33.3)	>0.99
46–59 n (%)	12 (20)	12 (24)	0 (0)	0.24	1 (6.25)	0 (0)	>0.99
> 60 n (%)	11 (19)	10 (20)	1 (12)	0.62	1 (6.25)	1(33.3)	0.41
Glasgow Coma Scale (GSC)							
3–8 Severe n (%)	32 (54.2)	26 (51)	6 (75)	0.61	10 (62.5)	2 (67)	0.51
9–12 Moderate n (%) (Reference)	7 (11.8)	5 (10)	2 (25)		2 (12.5)	1 (33)	
13–15 Mild n (%)	20 (34)	20 (39)	0 (0)	0.05	4 (25)	0 (0)	0.43
Neurosurgery							
n (%)	30 (51)	24 (47)	6 (75)	0.25	9 (56)	2 (67)	>0.99
Acute symptomatic seizures (ASS)							
n (%)	13 (22)	11 (22)	2 (25)	>0.99	4 (25)	1 (33)	>0.99
Polytrauma							
n (%)	45 (76)	40 (78)	5 (63)	0.38	14 (87.50)	3 (100)	>0.99
Subarachnoid hemorrhage							
n (%)	30 (51)	25 (51)	5 (62.50)	0.70	8 (50)	1 (33)	>0.99
Epidural hematoma							
n (%)	15 (25)	13 (25)	2 (25)	>0.99	3 (19)	1 (33)	0.53
Subdural hematoma							
n (%)	27 (46)	23 (45)	4 (50)	>0.99	9 (56)	3 (100)	0.26
Midline shift							
n (%)	16 (27)	11 (22)	5 (62.50)	0.02	4 (25)	1 (33)	>0.99
Fracture							
n (%)	21 (36)	16 (31)	5 (62.50)	0.70	12 (75)	0 (0)	0.03
Mortality							
Alive n (%)	52 (88)	44 (86)	8 (100)	0.57	14 (87.50)	3 (100)	>0.99
Death n (%)	7 (12)	7 (14)	0 (0)		2 (12.50)	0 (0)	
Family history of epilepsy							
No n (%)	36 (61)	31 (61)	5 (62.5)	0.66	12 (75)	3 (100)	>0.99
Yes n (%) (Reference)	11 (19)	9 (18)	2 (25)		2 (12.5)	0 (0)	
Maybe n (%)	1 (1)	1 (1)	0 (0)	>0.99	0 (0)	0 (0)	>0.99
Not informed n (%)	11 (19)	10 (20)	1 (12.5)	>0.99	2 (12.5)	0 (0)	>0.99
Quantity of lesions							
No lesion n (%)	1 (1)	1 (2)	0 (0)	>0.99	1 (6)	0 (0)	>0.99
One lesion n (%) (Reference)	27 (46)	23 (45)	4 (50)		7 (44)	1 (33)	
Two lesions n (%)	11 (19)	10 (20)	1 (12.5)	>0.99	2 (12.5)	0 (0)	>0.99
Multiple lesions n (%)	20 (34)	17 (33)	3 (37.5)	>0.99	6 (37.5)	2 (67)	>0.99
Localization of the lesions							
Frontal							
With lesion n (%)	40 (68)	16 (31)	3 (37)	0.70	10 (62.50)	1 (33)	0.54
Temporal							
With lesion n (%)	31 (53)	25 (49)	6 (75)	0.26	9 (56.25)	3 (100)	0.26
Parietal							
With lesion n (%)	5 (8)	4 (8)	1 (12)	0.53	1 (6)	2 (67)	0.05
Occipital							
With lesion n (%)	1 (2)	1(2)	0(0)	>0.99	1 (6)	0 (0)	>0.99
Others							
With lesion n (%)	1 (2)	1 (2)	0 (0)	>0.99	0 (0)	0 (0)	>0.99
Lateralization of lesions							
No lesion n (%)	1 (2)	1 (2)	0 (0)	>0.99	1 (6.25)	0 (0)	>0.99
Unilateral n (%) (Reference)	42 (71)	35 (69)	7 (88)		10 (62.5)	2 (67)	
Bilateral n (%)	16 (27)	15 (29)	1 (12)	0.42	5 (31.25)	1(33)	>0.99
ApoE							
ε3/ε3 n (%) (Reference)	37 (63)	33 (65)	4(50)		11(69)	0(0)	
ε3/ε4 n (%)	16 (27)	12 (23)	4 (50)	0.22	4 (25)	3 (100)	0.04*
ε2/ε3 n (%)	3 (5)	3 (6)	0 (0)	>0.99	1 (6)	0 (0)	>0.99

(*continued on next page*)

Table 1 (*continued*)

Variables	Total population (n= 59)	Genetic Population (n=59)			Cytokine Population mRNA (n=19)		
		No-PTE (n=51)	PTE (n=8)	p-value	No-PTE (n=15)	PTE (n=4)	p-value
ε2/ε4 n (%)	1 (2)	1 (2)	0 (0)	>0.99	0(0)	0(0)	>0.99
ε4/ ε4 n (%)	2 (3)	2 (4)	0(0)	>0.99	0(0)	0(0)	>0.99
ε2/ε2 n (%)	0(0)	0(0)	0(0)	>0.99			
GAD 1- rs3791878							
*GG n (%) (Reference)	37 (63)	33(65)	4 (50)		9 (56)	2 (67)	
GT n (%)	20 (34)	16 (31)	4 (50)	0.43	6 (38)	1 (33)	>0.99
TT n (%)	2 (3)	2 (4)	0 (0)	>0.99	1 (6)	0 (0)	>0.99
GAD1- rs3828275							
CC n (%) (Reference)	19 (32)	15(29)	4(50)		7(44)	1(33.33)	
CT n (%)	30 (51)	27(53)	3 (38)	0.41	8(50)	1(33.33)	>0.99
*TT n (%)	10 (17)	9(18)	1(12)	0.63	1(6)	1(33.33)	0.37
GAD1 -rs769391							
*AA n (%) (Reference)	31 (53)	26(51)	5(63)		8(50)	3 (100)	
AG n (%)	26 (44)	23(45)	3(37)	0.71	8(50)	0 (0)	0.22
GG n (%)	2 (3)	2 (4)	0 (0)	>0.99	0 (0)	0 (0)	>0.99

Data for the total and subdivided population according to treatment with Non-PTE or PTE for genetic and cytokine analysis. Continuous variables were described as mean ± standard deviation (s.d), while categorical variables were described as distribution and frequency. (*) Represents alleles with increased risk for PTE occurrence as per Darrah report (Darrah et al., 2013). p-value significant differences are highlighted in bold.

Table 2
Association between TBI severity and presence of polytrauma, need of neurosurgery, occurrence of acute symptomatic seizure and development of post-traumatic epilepsy.

Features	Mild		Moderate & Severe		Odds Ratio		
	n	%	n	%	95 %CI	OR	*p-value*
Polytrauma	12	20.3	33	55.9	2.11–2.64	8.26	<0.05*
Neurosurgery	5	8.5	25	42.4	2.24–2.82	8.53	<0.05*
ASS	3	5.1	10	17	1.73–2.73	4.87	>0.05
PTE	0	0	8	13.6	2.36–3.13	7.57	<0.05*

For all variables, it is represented the n (%) of patients that presented the variable, considering the total n=59, except for #PTE that considered n=52, for excluding 7 patients that died or did not complete the 2-year follow-up period. Significant differences are highlighted in bold.

patients (5.31 ±0.33; see Fig. 2A) in the acute phase of TBI. Interestingly, patients with moderate and severe TBI also demonstrated a significant ($p<0.05$) higher *TNF-α* mRNA expression (5.88 ± 0.92) than the patients with mild TBI (5.13 ± 0.33; Fig. 2F). Absent values were excluded from the dataset due to their status as outliers.

4. Discussion

This study integratively assessed various characteristics of patients who were victims of acute TBI, regarding demographic, clinical, genetic, and inflammatory response profiles related to the development of PTE over 2 years. We confirmed the correlation between trauma severity and bilateral lesions with PTE likelihood. Genetic profile in our study revealed predominance of ε3 homozygosis in the ApoE gene analysis and higher occurrence of wild-type GG and AA in GAD1 SNPs rs3791878 and rs769391, respectively in the population of the present study. The mRNA analysis showed elevated levels of TNF-α in patients that later developed PTE.

Exploratory analysis identified two profiles: PTE and Non-PTE. PTE profile was more associated with severe TBI, presence of hematomas and bilateral lesions, and neurosurgery requirement.

Our study finding of a higher predominance of males under 45 years aligns with previous researchers, suggesting it may be linked to factors like reckless behavior in traffic by men, as well as falls related to work activities and risk sports, among other reasons (Freitas, Ribeiro and Jorge, 2007; Xu et al., 2017).

Despite a relatively low PTE occurrence (8/59, 13.5 %) in our 2-year follow-up study compared to other research (Englander et al., 2003), our exploratory analysis showed that patients associated with PTE development exhibited specific profiles, such as severe TBI, bilateral lesions, lesions in the temporal and parietal regions, polytraumas, hematomas, and hemorrhage, and requiring neurosurgical intervention. While these characteristics have been individually described by other studies (Englander et al., 2003; Tube et al., 2019; Foresti et al., 2023), our study presents them as a group of variables that collectively characterize a profile.

Regarding ApoE genotype analysis, our study found that most patients had the allele ε3 in homozygosis, while few of them presented the allele ε4 in homozygosis. These facts align with global allele distribution, were the ε3 allele is the most common (78 %), followed by ε4 (14 %), considered harmful, and ε2 (8 %), a potential neuroprotective (Singh, Singh and Mastana, 2006; Lumsden et al., 2020). While some authors have reported an increased incidence of ApoE4 alleles in patients with temporal lobe epilepsy (TLE) and refractory epilepsy within the general population (Fu et al., 2010; Gong et al., 2017), other studies have reported conflicting findings, as no significant correlation between ApoE and EPT (Cavalleri et al., 2005; Anderson et al., 2009; Raymont et al., 2010; Misra *et al.*, 2023), and our study provides a robust contribution for it.

Our exploratory analysis of the GAD 1 SNP genotype associated specific genotypes (GG for SNP rs3791878, CT for SNP rs3828275, and AA for SNP rs769391 with the PTE profile. Initial studies of genetic variability in the A1AR with posttraumatic seizures demonstrated the GG genotype for rs3766553 was associated with increased late post-traumatic seizure rates. Also, the CT genotype for SNP rs10920573 revealed an association with increased late posttraumatic seizures. Multiple risk genotype analysis showed subjects with both risk genotypes had a 46.7 % chance of late posttraumatic seizures (Wagner et al., 2010). Darrah and colleagues reported as well that the homozygosis wild type GG (rs3791878) and AA (rs769391) were associated with PTE after TBI. The authors also mentioned that having both variants increases the risk probability of having posttraumatic seizures (Darrah et al., 2013). Despite not identifying any significant GAD1 SNP risk factors in our study, possibly attributed to the small sample size of PTE, the MCA analysis proved instrumental in visualizing relationships between categorical variables.

Concerning inflammatory factors, an increased acute mRNA expression of TNF-α was observed in patients with moderate or severe trauma and specially in patients who later developed PTE. This aligns

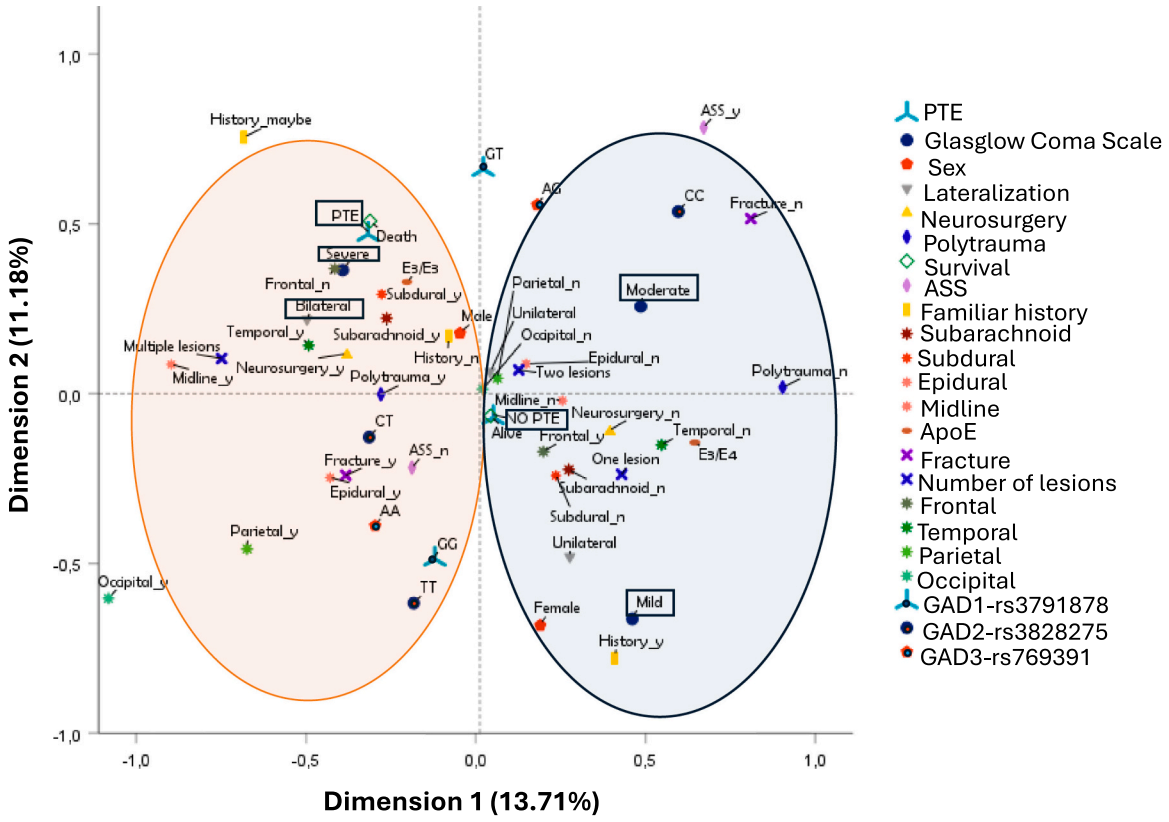


Fig. 1. Multiple Correspondence Analysis (MCA). Associations of Dimensions 1 (Dimension 1; 13.7 % of the variance) and 2 (Dimension 2; 11.18 % of the variance) of MCA in relation to Posttraumatic Epilepsy (PTE), survival, sex, Glasgow Coma Scale (GCS) on hospital admission, need for neurosurgery n=59.Acute Symptomatic Seizure (ASS), polytrauma, family history of epilepsy, quantity of lesions, lateralization of lesions, location of lesions (Frontal, Temporal, Parietal, Occipital), presence of fracture, subarachnoid hemorrhage, subdural and epidural hematoma, midline shift, ApoE genotype, and GAD1 SNP1 (rs3828275), SNP2 (rs3791878), SNP3 (rs769391). Joint category plots of the explored variables. The orange circle represents the profile called PTE, and the blue circle represents the profile called Non-PTE. Variables with an underscore followed by the letter "y" (e.g., Polytrauma_y) indicate that the variable is present, while variables with an underscore followed by the letter "n" (e.g., Polytrauma_n) indicate that the variable is not present. GCS is described as one of three categories, Mild, Moderate and Severe. Reference to additional notations in the figure might be gathered by the symbol code on the inset and their corresponding match on the figure.

with other studies showing increase serum and CSF, TNF-α levels in patients with epilepsy (Li et al., 2018; Qiu et al., 2021). Additionally, patients with refractory epilepsy also have higher serum TNF-α, and this increased expression suggests an association with epilepsy severity (Soltani Khaboushan et al., 2022).

After trauma, microglial cells are activated and release several inflammatory mediators, one of them being TNF-α, that may promote the immune response to TBI, increasing the permeability of the blood-brain barrier and aiding in the recruitment of peripheral immune cells. A study by Scherbel and colleagues in mice knockout for TNF-α demonstrated that TNF-α produced in the acute early stage of TBI can be deleterious; however, the absence of this cytokine can increase neuronal loss and recovery time during the chronic period (Scherbel et al., 1999). Our study is the first to indicate early levels of TNF- α after TBI with the later onset of PTE. The search for biomarkers for PTE is a hot area (Arulsamy and Shaikh, 2020; Sun et al., 2021) and our study provides robust evidence on this.

Nevertheless, TNF-α efficacy as a sole biomarker is limited by the confounding effect of injury severity on the TNF-α elevation-PTE relationship. Its influence on the post-TBI development and recovery process remains unclear and intricate Its ubiquitous expression may also complicate its utility when applied in the context of other disease states (Casault et al., 2019).

Inherent to our study are several limitations that warrant acknowledgment. Firstly, the sample size was relatively small, which may limit the generalization of our findings to broader populations. Additionally, due to resource and logistic constraints, certain outcome measures, such as healthy controls, baseline (time zero) sample collection and long-term analysis of cytokines, were not included in our analysis, potentially restricting the comprehensiveness of our conclusions. Despite these limitations, and the unpredictable study's challenges posed by the COVID-19 pandemic, including the direct impact on patient enrollment and follow-up schedules due to social distancing measures imposed between 2020 and 2021, it pioneered the concomitant examination of GAD, ApoE and inflammatory response in TBI patients, emphasizing the need for continued research with expanded patient cohorts to consolidate results regarding PTE development in these patients. A current clinical trial, with adjusted protocol, is currently underway (Foresti et al., 2002).

Furthermore, it is unlikely that any single factor might in isolation have a strong causative influence over the development of epilepsy after TBI. Our results provide a suggestion of potential clusterings that might be relevant as prognostic factors for PTE. Future studies might try to assess the widest possible set of variables in order to shed light on the mechanistic aspects of epileptogenesis and PTE. Despite its high prevalence and association with adverse neurological outcomes, treatment strategies for PTE are variable and poorly defined. Promising research has identified clinical, EEG, imaging and fluid biomarkers that might aid stratification of PTE risk (Pease et al., 2024).

5. Conclusion

In summary our findings provide valuable insights into the intricate interplay of clinical, genetic, and inflammatory factors in TBI patients

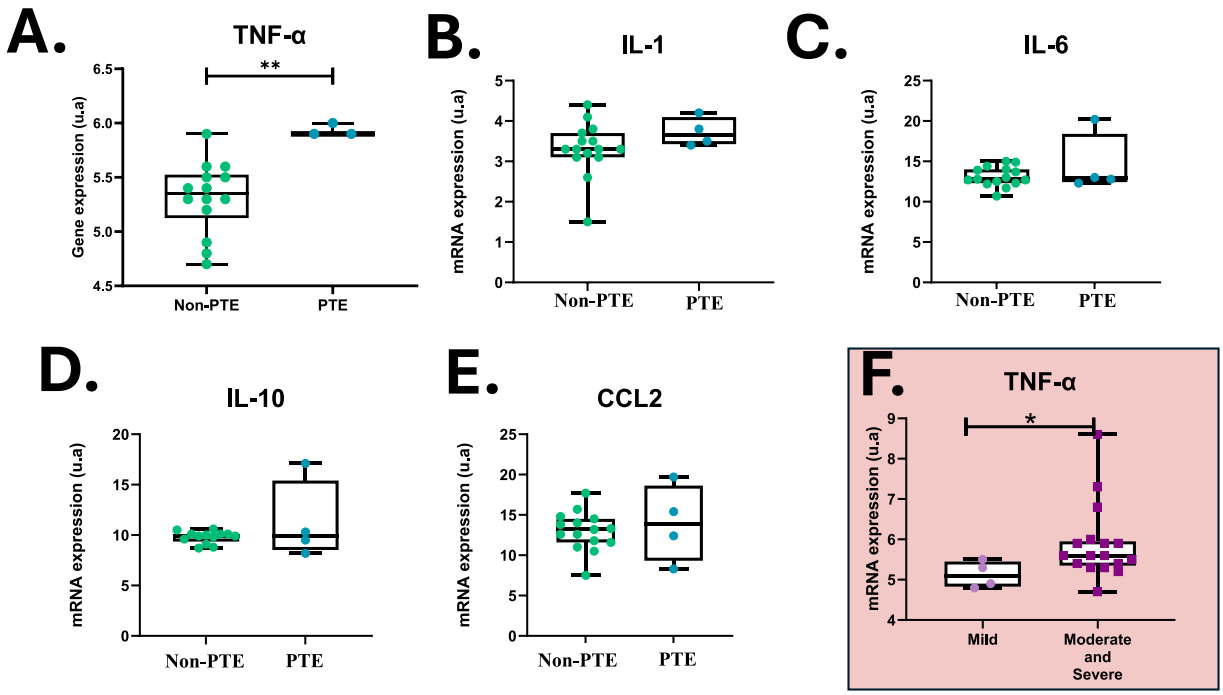


Fig. 2. mRNA expression of interleukins. **A.** mRNA expression of TNF-α between the Non-PTE and PTE groups. **B-E.** mRNA expression of IL-1, IL-6, IL-10, and CCL2 between the Non-PTE and PTE groups. **F.** mRNA expression of TNF-α between the mild and moderate-severer TBI groups. (Non-PTE =15, PTE = 4) *$p<0.05$, **$p<0.01$.

related to epileptogenesis, shedding light on potential avenues for further research and therapeutic exploration.

Ethics approval statement

The study protocol was approved by the Research Ethics Committee of Hospital das Clínicas da Faculdade de Medicina da Universidade de São Paulo (HC-FMUSP; CAAE n 08533513.6.2002.0068) and Universidade Federal de São Paulo (UNIFESP; CAAE n 08533513.6.1001.5505).

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Clara Pompeu: Writing – original draft, Data curation. **Priscila Nicolicht-Amorim:** Writing – original draft, Data curation. **Clivandir S. Silva:** Writing – original draft, Data curation. **Viviam Sanabria:** Writing – review & editing, Writing – original draft, Formal analysis, Data curation, Conceptualization. **Amanda C. Mosini:** Writing – review & editing, Writing – original draft, Formal analysis, Data curation, Conceptualization. **Michele L. Calió:** Writing – original draft, Data curation. **Thábatta K.E. Nakamura:** Writing – original draft, Data curation, Conceptualization. **Marimelia A. Porcionatto:** Writing – review & editing, Validation, Conceptualization. **Maria da Graça Naffah-Mazzacoratti:** Writing – review & editing, Writing – original draft, Validation, Project administration, Formal analysis, Conceptualization. **Maira Licia Foresti:** Writing – review & editing, Validation, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Luiz Eugênio Mello:** Writing – review & editing, Validation, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Conceptualization.

Declaration of Competing Interest

None of the authors has any conflict of interest to disclose.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.eplepsyres.2024.107402.

References

Agrawal, A., et al., 2006. Post-traumatic epilepsy: An overview. Clin. Neurol. Neurosurg. 108 (5), 433–439. https://doi.org/10.1016/j.clineuro.2005.09.001.

Anderson, G.D., et al., 2009. Haptoglobin phenotype and apolipoprotein E polymorphism: Relationship to posttraumatic seizures and neuropsychological functioning after traumatic brain injury. Epilepsy Behav. 16 (3), 501–506. https://doi.org/10.1016/j.yebeh.2009.08.025.

Arulsamy, A., Shaikh, M.F., 2020. Tumor Necrosis Factor-α, the Pathological Key to Post-Traumatic Epilepsy: A Comprehensive Systematic Review. ACS Chem. Neurosci. 11 (13), 1900–1908. https://doi.org/10.1021/acschemneuro.0c00301.

Ayele, D., Zewotir, T., Mwambi, H., 2014. Multiple correspondence analysis as a tool for analysis of large health surveys in African settings. Afr. Health Sci. 14 (4).

Baki, C.B., et al., 2022. Assessment of Hydro-Agricultural Infrastructures in Burkina Faso by Using Multiple Correspondence Analysis Approach. Sustainability 14.

Casault, C., et al., 2019. Cytokine responses in severe traumatic brain injury: where there is smoke, is there fire ? Neurocritical Care 30 (1), 22–32. https://doi.org/10.1007/s12028-018-0522-z.

Cavalleri, G.L., et al., 2005. Failure to replicate previously reported genetic associations with sporadic temporal lobe epilepsy: Where to from here? Brain 128 (8), 1832–1840. https://doi.org/10.1093/brain/awh524.

Costa, P.S., et al., 2013. The use of multiple correspondence analysis to explore associations between categories of qualitative variables in healthy ageing. J. Aging Res. 2013.

Cotter, D., Kelso, A., Neligan, A., 2017. Genetic biomarkers of posttraumatic epilepsy: A systematic review. Seizure 46, 53–58. https://doi.org/10.1016/j.seizure.2017.02.002.

Cronbach, L.J., 1951. Coefficient alpha and the internal structure of test. Psychometrika 16, 297–334.

Darrah, S.D., et al., 2013. Genetic variability in glutamic acid decarboxylase genes: Associations with post-traumatic seizures after severe TBI. Epilepsy Res. 103 (2–3), 180–194. https://doi.org/10.1016/j.eplepsyres.2012.07.006.

Englander, J., et al., 2003. Analyzing risk factors for late posttraumatic seizures: A prospective, multicenter investigation. Arch. Phys. Med. Rehabil. 84 (3 SUPPL. 1), 365–373. https://doi.org/10.1053/apmr.2003.50022.

Foresti, M.L., et al., 2023. Initial clinical evidence on biperiden as antiepileptogenic after traumatic brain injury. A Randomized Clinical Trial Running Title: Use of Biperiden for the Prevention of Post-traumatic Epilepsy Maira Licia Foresti (Available at). *medRxiv*, ((Pre-Print.). https://doi.org/10.1101/2023.11.10.23298341.

Freitas, J.P.P., Ribeiro, L.A., Jorge, M.T., 2007. Pediatric victms of traffic accidents admitted to a university hospital: Epidemiological and clinical aspects. Cad. De. Saude Publica 23 (12), 3055–3060. https://doi.org/10.1590/s0102-311x2007001200028.

French, J.A., et al., 2021. Antiepileptogenesis and disease modification: Clinical and regulatory issues. Epilepsia Open 6 (3), 483–492. https://doi.org/10.1002/epi4.12526.

Fu, Y. hui, et al., 2010. Association of apolipoprotein E polymorphisms with temporal lobe epilepsy in a Chinese Han population. Epilepsy Res. 91 (2–3), 253–259. https://doi.org/10.1016/j.eplepsyres.2010.07.020.

Gong, J.E., et al., 2017. Common variants of APOE are associated with anti-epileptic drugs resistance in Han Chinese patients. Int. J. Neurosci. 127 (1), 14–19. https://doi.org/10.3109/00207454.2015.1137295.

Johnson, R.A., Wichern, D.W., 2007. *Applied Multivariate Statistical Analysis*. Sixth Edit. Pearson Prentice Hall, Upper Saddle River, New Jearsey, USA.

Karlander, M., Ljungqvist, J., Zelano, J., 2021. Post-traumatic epilepsy in adults: a nationwide register- based study. J. Neurol., Neurosurg., Psychiatry 92, 617–621. https://doi.org/10.1136/jnnp-2020-325382.

Khangar, N.V., Kamalja, K.K., 2017. Multiple Correspondence Analysis and its applications. Electron. J. Appl. Stat. Anal. 10, 432–462.

Li, T.R., et al., 2018. Correlation between tumor necrosis factor alpha mRNA and microRNA-155 expression in rat models and patients with temporal lobe epilepsy. Brain Res. 1700, 56–65. https://doi.org/10.1016/j.brainres.2018.07.013.

Lumsden, A.L., et al., 2020. Apolipoprotein E (APOE) genotype-associated disease risks: a phenome-wide, registry-based, case-control study utilising the UK Biobank. EBioMedicine 59. https://doi.org/10.1016/j.ebiom.2020.102954.

Maas, R, A.I., et al., 2017. Traumatic brain injury: Integrated approaches to improve prevention, clinical care, and research. Lancet Neurol. 16 (12), 987–1048. https://doi.org/10.1016/S1474-4422(17)30371-X.

Merritt, V.C., et al., 2021. Apolipoprotein E (APOE) ε4 Genotype Influences Memory Performance Following Remote Traumatic Brain Injury in U.S. Military Service Members and Veterans. Brain Cogn. 154, 105790.

Pease, M., et al., 2024. Insights into epileptogenesis from post-traumatic epilepsy. Nat. Rev. Neurol. 20 (May), 298–312. https://doi.org/10.1038/s41582-024-00954-y.

Qiu, L., et al., 2021. Relationship between tumor necrosis factor-alpha and neuropeptide y expression and neurological function score in epileptic children. Iran. J. Public Health 50 (5), 1056–1064. https://doi.org/10.18502/ijph.v50i5.6123.

Raymont, V., et al., 2010. Correlates of posttraumatic epilepsy 35 years following combat brain injury. Neurology 75 (3), 224–229. https://doi.org/10.1212/WNL.0b013e3181e8e6d0.

Sarlo, G.L., Holton, K.F., 2021. Brain concentrations of glutamate and GABA in human epilepsy: A review. Seizure 91 (January)), 213–227. https://doi.org/10.1016/j.seizure.2021.06.028.

Scherbel, U., et al., 1999. Differential acute and chronic responses of tumor necrosis factor-deficient mice to experimental brain injury. Proc. Natl. Acad. Sci. USA 96 (15), 8721–8726. https://doi.org/10.1073/pnas.96.15.8721.

Singh, P.P., Singh, M., Mastana, S.S., 2006. APOE distribution in world populations with new data from India and the UK. Ann. Hum. Biol. 33 (3), 279–308. https://doi.org/10.1080/03014460600594513.

Soltani Khaboushan, A., Yazdanpanah, N., Rezaei, N., 2022. Neuroinflammation and Proinflammatory Cytokines in Epileptogenesis. Mol. Neurobiol. 59, 1724–1743.

Sun, L., et al., 2021. The Role of Neuroinflammation in Post-traumatic Epilepsy. Front. Neurol. 12 (May)), 1–13. https://doi.org/10.3389/fneur.2021.646152.

Walsh, P.S., Metzger, D.A., Higuchi, R., 2013. Biotechniques 30th anniversary gem Chelex 100 as a medium for simple extraction of DNA for PCR-based typing from forensic material. BioTechniques 54 (3), 506–513.

Woodcock, T., Morganti-Kossmann, M.C., 2013. The role of markers of inflammation in traumatic brain injury. Front. Neurol. 4 (MAR(March)), 1–18. https://doi.org/10.3389/fneur.2013.00018.

Xu, T., et al., 2017. Risk factors for posttraumatic epilepsy: A systematic review and meta-analysis. Epilepsy Behav. 67, 1–6. https://doi.org/10.1016/j.yebeh.2016.10.026.